Detailed Project Report

Factors Affecting Crime

Section A – Group 15

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1 Abstract

This project investigates the patterns, context, and operational factors associated with crime incidents using a cleaned Los Angeles crime dataset. The work combines data extraction, cleaning, exploratory data analysis (EDA), statistical testing, and machine learning experiments, and culminates in dashboard-ready KPI preparation. The objective is to identify when, where, and under what victim/context conditions crime concentration is highest, then convert those findings into actionable outputs for monitoring and decision support.

2 Project Objectives

- Build a reliable end-to-end analytics pipeline from raw data to reporting-ready outputs.
- Identify temporal, spatial, and victim-context factors linked with crime volume and concentration.
- Validate patterns through statistical methods (association tests, distribution checks, significance tests).
- Build baseline predictive models for binary/multiclass crime-related targets.
- Prepare aggregated indicators and visual artifacts suitable for dashboarding and presentation.

3 Repository Overview

The project is organized into a notebook-driven analytical workflow with supporting documentation and reports:

- `data/processed/crime_dataset_clean.csv`: Canonical cleaned dataset.
- `notebooks/01_extraction.ipynb`: Raw load and initial inspection.
- `notebooks/02_cleaning.ipynb`: Data quality treatment and cleaning transformations.
- `notebooks/eda.ipynb`: Univariate and bivariate exploratory analysis.
- `notebooks/Statistical_analysis.ipynb`: Statistical tests and ML experiments.
- `notebooks/final_load_prep.ipynb`: Final KPI/aggregation preparation.
- `docs/data_dictionary.md`: Detailed schema, lineage, and derived-feature definitions.
- `tableau/`: Dashboard integration area.

- `reports/`: Final report and presentation deliverables.

4 Data Summary

4.1 Canonical Dataset Profile

From the cleaned canonical file (`data/processed/crime_dataset_clean.csv`), the dataset profile is:

- Total records: **326,327**
- Total columns retained after cleaning: **24**
- Time span: **2020-01-01** to **2024-12-30**
- Spatial coverage: **21 area names**
- Unit of analysis: one row per incident report

4.2 Core Variables

The cleaned dataset contains incident identity/time fields, geography fields, crime class descriptors, victim attributes, and case disposition fields. Key examples include: `DR_NO`, `DATE OCC`, `TIME OCC`, `AREA NAME`, `Crm Cd Desc`, `Vict Age`, `Vict Sex`, `Vict Descent`, `Status Desc`, `LAT`, `LON`.

5 Data Pipeline and Processing

5.1 Stage 1: Extraction (`01_extraction.ipynb`)

- Loads the raw crime CSV and performs initial schema and shape inspection.
- Documents basic initial data quality issues for downstream treatment.

5.2 Stage 2: Cleaning (`02_cleaning.ipynb`)

Main cleaning actions observed:

- Date and time standardization: conversion of report/occurrence dates and formatting `TIME OCC`.
- De-duplication based on `DR_NO`.
- Category normalization in victim fields (e.g., unknown/unmapped categories normalized).

- Invalid victim age handling (≤ 0 corrected using median-imputation strategy).
- Sparse-column dropping (`Crm Cd 2/3/4`, `Cross Street`) due to high missingness.
- Final null handling and export of cleaned data.

5.3 Stage 3: EDA (`eda.ipynb`)

The EDA notebook performs:

- Univariate analysis for major numerical and categorical dimensions.
- Outlier detection and distribution-level inspection.
- Bivariate analysis for hour/day/month/area interactions with crime counts and categories.

5.4 Stage 4: Statistical and Modeling Analysis (`Statistical_analysis.ipynb`

This stage includes:

- Descriptive statistics (mean/median/mode, variance, skewness/kurtosis).
- Relationship analysis (correlation, cross-tabs, Cramer's V).
- Significance testing (Chi-square, ANOVA, t-test).
- Seasonal decomposition and anomaly-support metrics.
- Baseline machine learning models for classification and regression tasks.

5.5 Stage 5: Final KPI Prep (`final_load_prep.ipynb`)

The final notebook prepares dashboard-facing analytics:

- Volume KPIs (total crimes, growth measures).
- Temporal KPIs (peak-hour, night ratio, weekend ratio).
- Spatial concentration KPIs (hotspot area concentration).
- Category and enforcement-context readiness tables.

6 Key Empirical Findings from Cleaned Data

6.1 Area Concentration

Top-5 areas by volume in the cleaned dataset:

1. 77th Street – 28,932 incidents
2. Central – 24,560 incidents
3. Southeast – 23,024 incidents
4. Southwest – 20,539 incidents
5. Hollywood – 19,694 incidents

These results indicate clear geographic concentration and justify area-wise prioritization in interventions.

6.2 Most Frequent Crime Categories

Top categories by count:

1. Battery – Simple Assault (74,704)
2. Assault with Deadly Weapon, Aggravated Assault (53,451)
3. Intimate Partner – Simple Assault (46,583)
4. Robbery (32,219)
5. Criminal Threats – No Weapon Displayed (19,094)

Assault-linked offenses dominate the observed frequency distribution.

6.3 Victim Profile Snapshot

- Victim sex distribution: M (157,996), F (149,047), X (19,257), H (27).
- Mean victim age: 37.26 years; median victim age: 35 years.

6.4 Case and Severity Context

- Part distribution: Part 1 (148,474), Part 2 (177,853).
- Status descriptions are led by `Invest Cont` (190,391), followed by `Adult Other` and `Adult Arrest`.

6.5 Temporal Pattern Indicator

The monthly count peak appears in **July 2023** (7,208 incidents), while **December 2024** appears very low due to partial-period capture in the available extract. This emphasizes the need to treat terminal-month data carefully when interpreting trend extremes.

7 Derived Features and Analytical Utility

The notebooks construct several derived features for advanced analysis and modeling:

- Temporal: `hour, day, month, year`
- Victim segmentation: `Age Group`
- Binary target: `is_violent`
- Multiclass target: `Crm_Type_Encoded`
- Trend fields: `Crime_Count, Ordinal_Date, MoM_growth, YoY_growth, Z_Score`

These transformations convert raw incident records into interpretable risk and monitoring indicators.

8 Statistical and Machine Learning Components

8.1 Statistical Tests

Implemented analyses include:

- Chi-square test for categorical association.
- ANOVA and t-tests for comparative inferential assessment.
- Correlation and cross-tab analyses for dependency exploration.
- Seasonal decomposition and z-score based anomaly context.

8.2 Machine Learning Experiments

Implemented model families:

- Logistic Regression (binary classification context).
- Random Forest Classifier (multiclass context).
- Linear Regression (time-aggregated crime trend context).

Evaluation utilities in the notebook include `classification_report`, `accuracy_score`, and regression metrics (`MSE`, R^2).

9 Dashboard Integration (Tableau)

The project dashboard is available at:
https://public.tableau.com/app/profile/apoorva.choudhary3398/viz/Factors_Affecting_

`Crime/VictimContextandEnforcement?publish=yes`

9.1 Embedded Dashboard Visuals

The repository contains dashboard image exports integrated below.

Figure 1: Dashboard View 1: Victim Context and Enforcement (Tableau export)

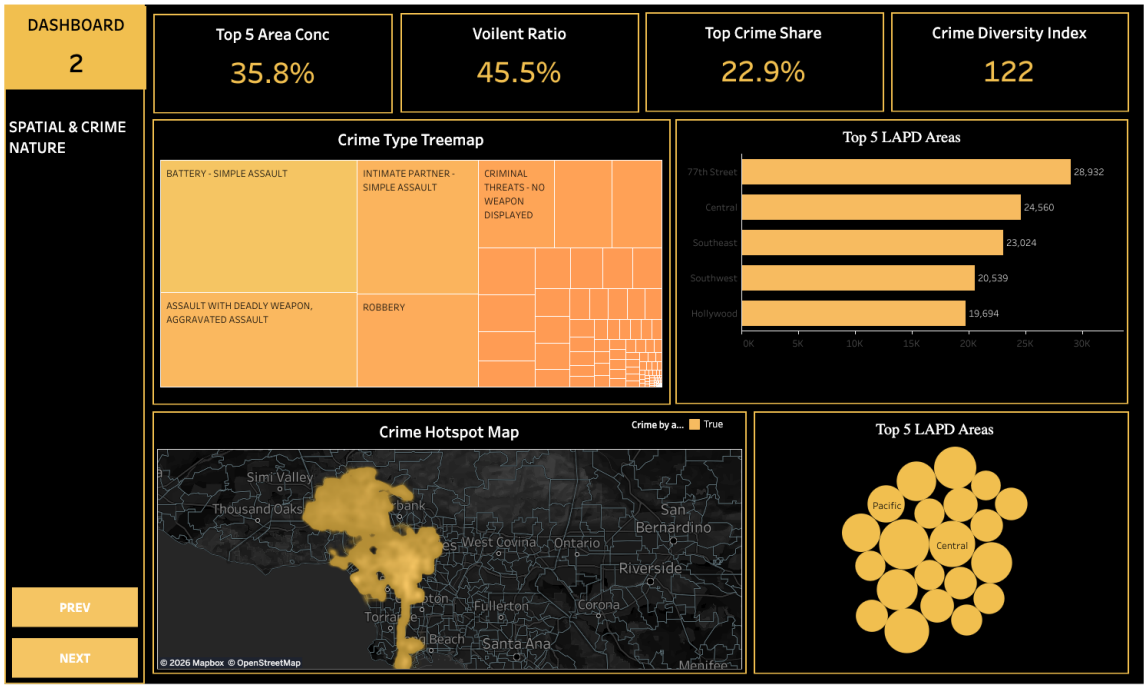

Figure 2: Dashboard View 2: Crime Pattern and Context Indicators (Tableau export)

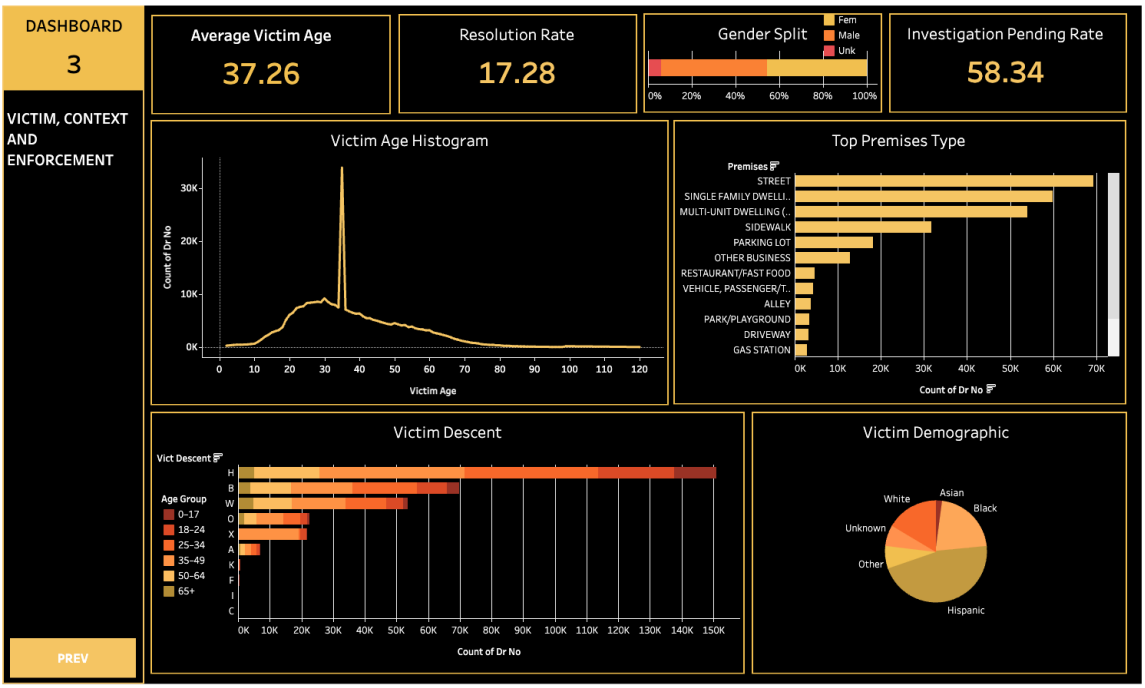

Figure 3: Dashboard View 3: Comparative and Segment-level Perspective (Tableau export)

10 Tools, Libraries, and Execution Notes

10.1 Primary Libraries Referenced in Notebooks

- Data handling: `pandas`, `numpy`
- Visualization: `matplotlib`, `seaborn`
- Statistics: `scipy`, `statsmodels`
- ML: `scikit-learn`

10.2 Execution Sequence

Recommended notebook run order:

1. `01_extraction.ipynb`
2. `02_cleaning.ipynb`
3. `eda.ipynb`
4. `Statistical_analysis.ipynb`
5. `final_load_prep.ipynb`

10.3 Environment Notes

Some notebook paths indicate Colab-style execution (`/content/...`). For reproducible local execution, path harmonization and explicit dependency pinning are recommended.

11 Limitations

- The repository currently lacks a finalized dependency manifest (`requirements.txt`/`environment.yn`
- Some data path names differ across notebooks and should be unified for strict reproducibility.
- A terminal month with low count likely reflects partial-period data and can bias naive trend comparisons.

12 Conclusions

The project establishes a complete analytical workflow from raw crime records to dashboard-ready insights. The strongest empirical signals emerge from (a) geographic concentration in selected divisions, (b) high prevalence of assault-linked categories, and (c) measurable temporal concentration pockets. Statistical testing and baseline ML models complement the EDA layer by adding inferential and predictive framing. The resulting outputs are suitable for periodic monitoring, enforcement planning support, and future model refinement.

13 Future Work

- Add fully reproducible environment configuration and one-click pipeline orchestration.
- Expand geospatial analysis by robust handling of invalid coordinates and mapping layers.
- Introduce advanced forecasting/classification baselines with systematic hyperparameter tuning.
- Integrate automated data quality tests and CI checks for notebook outputs.
- Connect Tableau directly to refreshed processed tables for near-real-time updates.

14 References

1. Project Data Dictionary: `docs/data_dictionary.md`

2. Tableau Dashboard: https://public.tableau.com/app/profile/apoorva.choudhary3398/viz/Factors_Affecting_Crime/VictimContextandEnforcement?publish=yes
3. Canonical processed data: `data/processed/crime_dataset_clean.csv`